



Sharif University of Technology
Department of Electrical Engineering

Sensing and Measurement (25769)

Report of Course Project: Conditioning Circuit Design Problem

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Abstract

This report presents the design and simulation of a signal conditioning circuit for piezoelectric sensors intended for use in robotic hand motion replication. Piezoelectric transducers were selected due to their high sensitivity, fast response, and suitability for detecting dynamic mechanical deformation. The piezoelectric element was modeled using the Butterworth–Van Dyke (BVD) equivalent circuit to accurately represent its electrical and mechanical behavior. A multi-stage signal conditioning system was designed, consisting of high-impedance buffering, high-pass filtering for drift removal, differential amplification, and noise suppression filtering. The operational amplifier selection was based on low input bias current, rail-to-rail operation, and compatibility with low-voltage supply constraints. The complete system was implemented and analyzed using LTspice simulation software. Simulation results demonstrate that the circuit successfully amplifies low-amplitude piezoelectric signals into usable voltage levels while maintaining signal integrity and stability. The effects of filtering, amplification, and phase differences between sensor signals were evaluated. Furthermore, there is a discussion on potential sources of non-ideal behavior of the circuit and suggestions on how to solve or improve the situation in those cases. The results confirm that the designed conditioning circuit is suitable for motion sensing applications in robotic hand control systems.

Keywords: Conditioning Circuit, Piezoelectric Transducer, LTspice CAD Tool, Sensing and Measurement

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1 Component Selection

1.1 Transducer Selection

The nature of the measurement in this application is one of force, vibration and strain. Since the grip of the robotic hand consists of many micro-vibrations, and the primary challenge assigned is accounting for the tremor in the robotic hand at the end of a rotational movement, the dominant characteristic of the changes to measured, is the high frequency. The frequency chosen in the CAD simulation was 250Hz, however, the circuit has shown to work with high efficiency for frequencies as high as 800Hz as well.

Other criteria affecting the choice of transducer are high sensitivity, as the mechanical disturbances involved produce relatively small electrical signals, and physical constraints such as little space for installation.

Most typical mechanical transducers/sensors are as following:

- **Strain Gauges:** Having to be placed inside a Wheatstone bridge ([2]), they are passive sensors and thus need voltage supply. Also, their high sensitivity to temperature introduces measurement uncertainty, approximated by the relation below as stated in [3]:

$$GF = \frac{T_0}{T} GF_0 + C_1 \left(\frac{T_0}{T} \right)^2 \varepsilon$$

- **Resistive force sensors:** These sensors exhibit nonlinear response characteristics and limited repeatability, making them unsuitable for precise measurement applications ([1]).
- **Capacitive sensors:** Using such sensors would make the task of signal conditioning over-complicated.
- **MEMS accelerometers:** These are attractive due to low cost, small size, and integrated electronics, but their performance is limited by higher noise levels and reduced bandwidth compared to piezoelectric sensors, making them less suitable for accurate measurement of high-frequency mechanical disturbances.
- **Piezoelectric transducers:** Piezoelectric sensors generate electrical charge directly in response to mechanical stress and therefore do not require external excitation. They offer high sensitivity, wide bandwidth, fast dynamic response, and compact physical dimensions. These characteristics make them particularly suitable for detecting vibration and tremor in robotic hand applications.

Considering all requirements, piezoelectric sensing represents the most appropriate solution. The selected component is the **Murata 7BB-20-6 piezoelectric diaphragm** ([5]), which offers suitable electrical character-

istics, compact size, and sufficient bandwidth for detecting the vibration frequencies relevant to this application.

1.2 Amplifier Selection

The electrical signals generated by piezoelectric transducers are typically low in amplitude and originate from a high-impedance source. Therefore, proper amplification is essential to ensure accurate signal acquisition while preserving signal integrity.

The primary requirements for the amplifier in this application are summarized as follows:

- **High input impedance:** Piezoelectric sensors behave electrically as capacitive sources with very high output impedance. If the amplifier input impedance is insufficiently high, significant signal attenuation may occur due to loading effects. Therefore, the amplifier must provide extremely high input impedance to preserve the signal amplitude.
- **Low input bias current:** Input bias current flowing into the amplifier input produces voltage drops across the sensor's source impedance, resulting in measurement error and signal distortion. Minimizing input bias current is therefore essential for accurate signal conditioning.
- **Low noise performance:** Since the sensor output signals are small, amplifier noise can significantly degrade signal quality and reduce measurement accuracy. A low-noise amplifier ensures a higher signal-to-noise ratio and more reliable signal reproduction.
- **Adequate bandwidth:** The amplifier must support the frequency range of interest without introducing attenuation or distortion.

Based on these requirements, **TI's LMP7721 operational amplifier** ([7]) was selected for this application. This amplifier offers ultra-low input bias current in the femto-ampere range, extremely high input impedance, and low noise characteristics. These features make it particularly well-suited for signal conditioning of piezoelectric sensors. Additionally, its stability and precision characteristics ensure reliable amplification of low-amplitude vibration signals.

2 Conditioning Circuit Design

2.1 Modeling the Piezoelectric: the BVD Model

The next problem to discuss after choosing a transducer is how to model it in the CAD tool to perform the analysis with sufficient accuracy. Among the electrical equivalent circuits available for a piezoelectric, the Butterworth

Van Dyke model (BVD) seems to be of reasonable conformity to reality and yet enough simplicity.

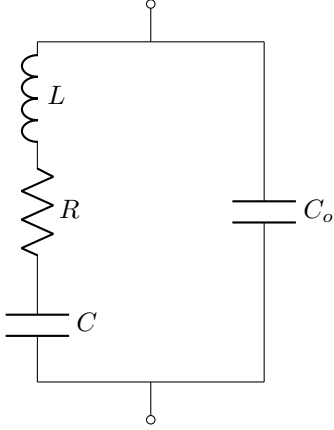


Figure 1: The Butterworth Van Dyke (BVD) model

In this model, C_0 is the clamped (static) capacitance, C_1 is the effective compliance, R_1 represents the mechanical and dielectric losses in total and L_1 is the effective mass. Now we can determine the component values using the component's data sheet [5] and the arguments in [4].

The data sheet gives the resonance frequency $f_s = 6.3kHz$, the resonant frequency which is equivalent to R in the model ($R = 350\Omega$) and the capacitance ($10nF$) which is equivalent to C_0 . Since we are not provided with the anti-resonance frequency f_p or a quality factor, one way is to assume a typical value for the *motional capacitance ratio* α in the relation $C_1 = \alpha C_0$. The typical range for α is 0.01 to 0.1, so 0.05 would be a good choice.

$$\alpha = 0.05 \rightarrow C_1 = (0.05)(10nF) = \boxed{0.5nF}$$

Obtaining the cyclic resonance frequency:

$$\omega_s = 2\pi f_s \rightarrow \omega_s \approx 2 \times 3.14 \times 6.3 \times 10^3 = 39.564 \times 10^3$$

Thus:

$$L_1 = \frac{1}{\omega_s^2 C_1} \approx \boxed{1.278H}$$

There are also more accurate methods of determining the circuit parameters, as argued in [6].

Now to complete the model we need to account for the charge generation property of the piezoelectric. We do so by connecting a current source parallel to the branches of the BVD model. This source indeed acts as the input to the conditioning circuit and represents the electrical equivalent of the mechanical excitation applied to the transducer. Since a piezoelectric element generates charge proportional to the applied strain, and therefore to the displacement amplitude in the linear operating region, the current source amplitude must reflect the expected mechanical vibration levels.

In this work, displacement amplitudes of $0.2 \mu m$ and $0.6 \mu m$ were selected as representative values within the

typical operating range of the device, ensuring operation well within the linear regime while allowing comparison between moderate and higher excitation conditions. The excitation frequency was set to $250 Hz$, which lies well below the transducer's resonance frequency ($\approx 6.3kHz$), so that the response remains dominated by the quasi-static capacitive behavior rather than resonance effects. This allows validation of the conditioning circuit and equivalent model under controlled conditions without introducing additional amplification or phase distortion associated with resonance.

At the last stage of the circuit, a simple DC stage is also added to give the resulting signal a desired DC offset.

2.2 Filters

The signal produced by the piezoelectric transducer and conditioning circuit contains the useful measurement component together with unwanted low-frequency drift, high-frequency noise, and power-line interference. Since the signal bandwidth of interest is well defined and relatively low, passive filters provide a simple, low-noise, and power-efficient solution. Passive implementations are particularly suitable here because they do not introduce additional active device noise, require no power supply, and do not risk saturation or bandwidth limitations associated with operational amplifiers. Furthermore, the high input impedance of the following amplification stage ensures minimal loading of the passive networks, preserving their designed frequency response.

2.2.1 HPF

The high-pass filter (HPF) is used to remove DC offsets and low-frequency drift originating from thermal effects, mechanical bias, and amplifier offset voltages. These components do not contain useful vibration information and may reduce the effective dynamic range of the system.

The cutoff frequency $f_c = 20Hz$ for instance, is sufficiently below the lowest frequency of interest ($250 Hz$) to avoid attenuation of the useful signal while still providing effective rejection of near-DC components.

$$f_c = \frac{1}{2\pi RC} \rightarrow R_H = 796k\Omega, C_H = 10nF$$

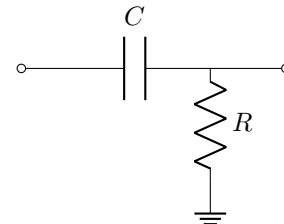


Figure 2: RC High Pass Filter Topology

2.2.2 LPF

The low-pass filter (LPF) is used to limit the bandwidth of the system and reduce high-frequency noise originating from electronic components, electromagnetic interference, and environmental sources. Since the useful signal is centered around 250 Hz and well below the mechanical resonance frequency of the transducer, higher-frequency components do not carry relevant measurement information.

The cutoff frequency $f_c = 1kHz$ may be a good choice to avoid attenuating the signal of interest while still providing attenuation of unwanted high-frequency noise. The same cutoff frequency relation applies:

$$f_c = \frac{1}{2\pi RC} \rightarrow R_L = 10k\Omega, C_L = 16nF$$

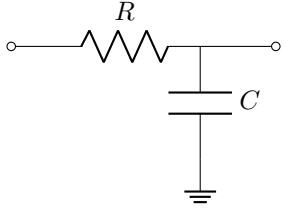


Figure 3: RC Low Pass Filter Topology

2.2.3 Notch Filter

The notch filter was implemented to suppress power-line interference at 50 Hz, which is commonly present due to electromagnetic coupling from mains-powered equipment. This interference can significantly affect measurement accuracy, particularly in low-amplitude signal conditions.

Calculating the component values:

$$f_c = \frac{1}{4\pi RC} \rightarrow R = 1k\Omega \text{ and } C = 1.6\mu F$$

Also another resistor R' with a rather large value was added to the structure to avoid the floating node issue in the CAD tool. Passive implementation is sufficient in this case because the objective is selective attenuation rather than amplification, and the signal amplitudes remain within a range where passive filtering provides adequate performance without introducing additional noise or complexity.

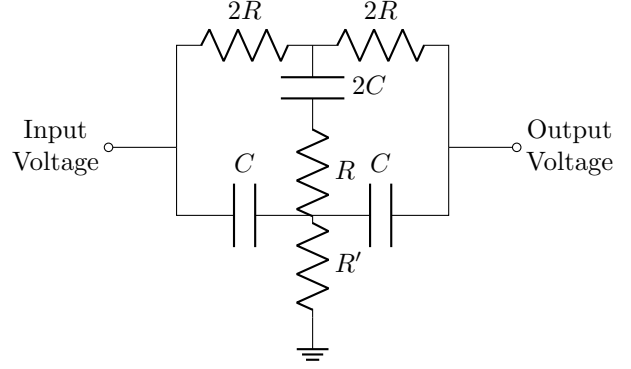


Figure 4: Notch Filter Topology

2.3 Amplification

To achieve the required signal level while maintaining stable and predictable frequency behavior, amplification was implemented using two cascaded active differential amplifier stages rather than a single high-gain instrumentation amplifier. Although instrumentation amplifiers offer excellent common-mode rejection and precision, applying very high gain in a single stage was observed in simulation to produce undesirable frequency-dependent effects, including bandwidth reduction and waveform distortion. The main culprit is possibly the fact that increasing the closed-loop gain reduces the effective bandwidth in circuits containing operational amplifiers.

To get a sense of the order of the input voltage, consider the first BVD model. The $0.2\mu A$ source is in parallel with the $10nF$ capacitor. So the absolute value of the voltage across the BVD structure is:

$$|V| = \frac{|I|}{\omega C_0}$$

where ω is $2\pi f$, f being the chosen value of $250Hz$, discussed before. From here, the input voltage is obtained to be approximately $13mV$.

Now since the amplifier is differential, the value to be amplified is even smaller. Two cascaded amplifier stages were assumed, each providing a gain of 100 to 110 (the two resistors on the left of Figure 5 being $110k$ and the other two, $1k$). This gain value was deliberately selected with additional margin to account for non-ideal behavior, component tolerances, and potential attenuation in preceding stages, ensuring the output signal remains within the desired voltage range.

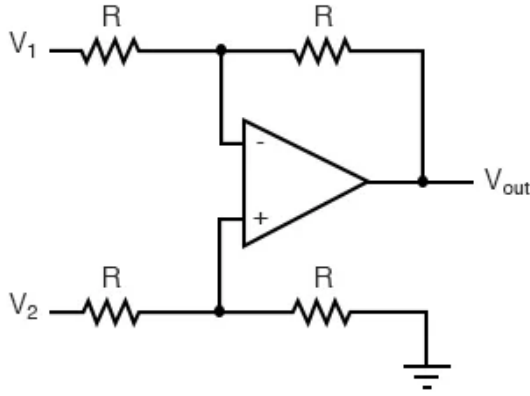


Figure 5: Active differential amplifier. Source: [All About Circuits](#)

2.4 LTspice Simulation Results

To verify the performance of the designed signal conditioning circuit, a complete simulation was conducted in LTspice using the piezoelectric transducer model, filtering stages, buffer, and cascaded differential amplification stages. The circuit designed in the graphic environment of LTspice is presented in Figure 6, which brings together all the parts discussed before.

The simulation results confirm that the conditioning circuit successfully amplifies the low-amplitude sensor signal while preserving waveform integrity. The final output waveform demonstrates stable amplification (no envelope drift or fluctuation), minimal distortion, and an output voltage level within the intended range of several hundred millivolts peak-to-peak (about 3.3V according to Figure 7). These results validate the effectiveness of the proposed circuit design for accurate sensing and conditioning of piezoelectric signals in robotic hand motion applications.

3 Discussion: Potential Causes of Abnormal Behavior

3.1 Assignment Summary

This section evaluates two critical signal anomalies—external noise interference and system parameter variations—affecting the robotic wrist’s vibration sensing circuit. By analyzing the current architecture, which includes high-precision LMP7721 amplifiers and specific filter stages, hardware-based solutions such as active filtering and DC-offset management are proposed to maintain signal integrity.

3.2 Switching Power Supply High-Frequency Noise (External Factor)

3.2.1 Description of the Factor

Robotic systems utilize motor drivers and DC-DC converters that generate high-frequency switching noise. Although the circuit features Low Pass Filters ($RL1, CL2$ and $RL2, CL3$) with a $10k\Omega$ resistance, these stages can still be bypassed by high-frequency electromagnetic interference (EMI) or power supply ripple entering through the $\pm 3.3V$ rails (VCC/VEE).

3.2.2 Effect on Signal

Signal Corruption: The output V_{out} will show high-frequency "spikes" or noise "grass" superimposed on the 250Hz vibration signal.

False Triggering: If the noise frequency aligns with the mechanical resonant frequency of the wrist, the control system may misinterpret electrical noise as actual physical oscillation, leading to unstable motor behavior.

3.2.3 Hardware Solution and Circuit Modification

Enhanced Decoupling: Add a $0.1\mu F$ ceramic capacitor in parallel with a $10\mu F$ electrolytic capacitor at the power pins of $U1$ through $U6$ to filter out rail-induced noise.

Active Low-Pass Stage: Currently, the Buffer stage ($U6$) is a simple voltage follower. By adding a feedback capacitor in parallel with a new feedback resistor, $U6$ can be converted into an active low-pass filter to provide sharper attenuation of frequencies above the 250Hz sensing range.

3.3 Thermal Drift and Sensor Aging (System Condition Change)

3.3.1 Description of the Factor

The piezoelectric model depends on sensitive components like $Rm1, Rm2$ (350Ω) and high-impedance bias resistors $RH1, RH2, RH3$ ($796k\Omega$). As the robot operates, internal heat changes the leakage current of the LMP7721 and the resistance values, causing the operating point of the Amp. Stages ($U3, U4$) to shift.

3.3.2 Effect on Signal

DC Offset Shift: The baseline of the vibration signal will "drift" away from the 0V reference. Even when the wrist is stationary, the V_{out} may show a constant DC voltage.

Dynamic Range Reduction: Because the Amp. Stages have high gain ($R6/R4 = 110k/1k = 110$), a small thermal drift at the input is amplified significantly. This can lead to signal clipping at the $\pm 3.3V$ rails, hiding the actual vibration peaks.

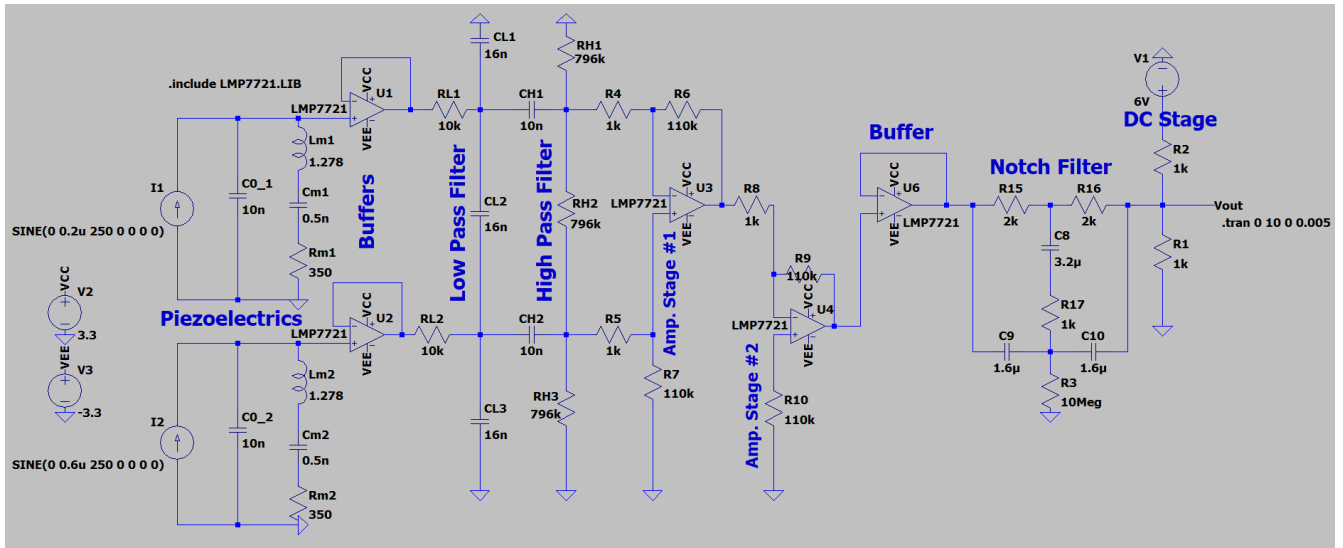


Figure 6: Circuit Designed in Ltspice's Graphic Environment

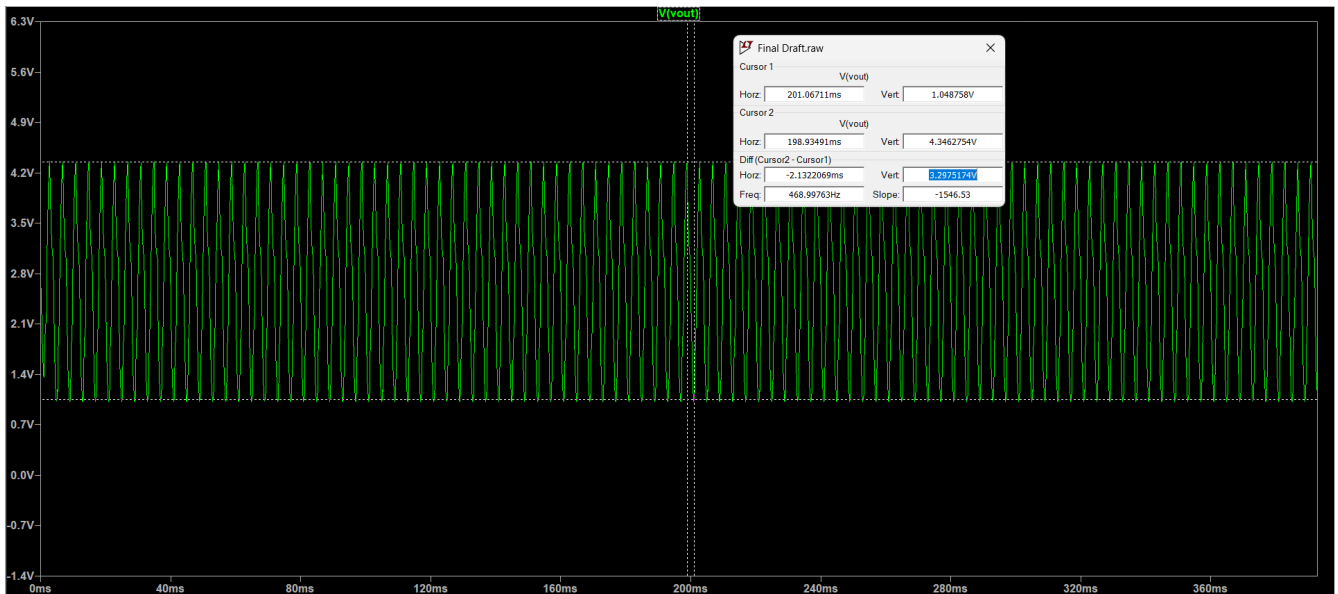


Figure 7: The Waveform of the Output Signal, Peak-to-Peak Can Be Read from the 'vert diff' Field

3.3.3 Hardware Solution Circuit Modification

DC Servo Integration: Install an active integrator circuit that samples V_{out} and feeds back a corrective DC signal to the non-inverting input of $U3$. This automatically "nulls" the drift in real-time.

Precision Component Matching: Replace the $110\text{k}\Omega$ and $1\text{k}\Omega$ gain resistors ($R4$ through $R10$) with high-precision, low-temperature-coefficient ($5\text{ppm}/^\circ\text{C}$) resistor networks. This ensures that the gain remains stable regardless of the robot's internal temperature rise.

4 Conclusion

In conclusion, the Butterworth–Van Dyke model provides a compact yet sufficiently accurate representation of the chosen piezoelectric sensor for this application, capturing both its static capacitive behavior and its resonant electromechanical dynamics. By extracting the motional parameters from the limited datasheet information and constraining them within typical ratios reported in the literature, the resulting equivalent circuit enables realistic LTspice simulations of the complete signal-conditioning chain. This modeling approach supports informed design decisions on front-end topology, gain, and bandwidth, while remaining flexible enough to be refined later with more detailed impedance measurements if required.

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